RESPONSE TO EDITOR AND REVIEWERS

Cogent Economics & Finance

Manuscript: AI Governance for Credit Decision Systems in P2P Lending

Decision: Major Revision (Revise and Resubmit)

Dear Dr McMillan, Deputy Academic Editor, and Reviewers,

We thank the editor and reviewers for the careful and constructive assessment of our manuscript. We have undertaken a substantial methodological and reporting revision rather than limiting the response to textual clarification. The revised manuscript now:

- defines outcome maturity using the April 2020 accounting cutoff and separates mature-cohort classification from a complementary competing-risk estimand that retains unresolved loans as right-censored;
- reconstructs the mature training cohort reproducibly and evaluates model families across 180,000, 360,000, 540,000, and the full 1,304,058-loan mature training sample;
- uses paired stratified bootstrap inference for model-performance differences and removes claims of model equivalence;
- recalibrates the validation-selected HistGB model using validation data only before repeating the threshold analysis;
- introduces explicit minimum-support requirements and uncertainty estimates for unemployment-subgroup and grade-by-term diagnostics;
- clarifies that the observed variation concerns analytical review rates, not loan renewal rates, and that no loan-renewal or refinancing process is modelled;
- publishes the analysis code and adds a reproducibility and policy-audit specification; and
- revises the governance framework, discussion, limitations, figures, and conclusion to distinguish empirical findings from regulatory or conceptual design implications and to avoid causal, equivalence, or unlawful-discrimination claims.

All responses below refer to the revised manuscript by section, subsection, table, or figure title rather than fixed page numbers because pagination may change during production.

Deputy Academic Editor / Reviewer 1

Comment 1. *Outcome maturity and sample selection: The restriction to outcome-mature loans retains only 10.3% of the test sample, potentially introducing selection bias. Clearly explain the maturity criteria and consider survival or competing-risk methods to account for right-censored observations. Provide robustness analyses to establish whether the principal findings remain valid.*

Response: We agree that conditioning the principal classification audit on outcome-mature loans changes the evaluation population and cannot be treated as if it removed censoring without consequence. We therefore made the maturity rule fully explicit and added a complementary time-to-event competing-risk analysis. Outcome maturity is now assessed through the April 2020 accounting cutoff. A loan is classified as mature if it has reached a terminal status (Fully Paid, Charged Off, or Default) by that cutoff, or if observed follow-up from origination to April 2020 is at least as long as its contractual 36- or 60-month term. Unresolved loans with shorter follow-up remain outcome-immature. We now state explicitly that this restriction may preferentially retain loans that resolve earlier and therefore defines a conditional estimand rather than a representative estimate for all originations.

The new competing-risk analysis retains unresolved loans as right-censored observations and treats full repayment, including early repayment, as a competing event to charge-off. Of 2,700,690 loans with usable timing, 175,144 are observed charge-offs, 765,339 are fully paid, and 1,760,207 remain right-censored. Charge-off cumulative incidence is 0.0413 at 12 months, 0.1061 at 24 months, and 0.1419 at 36 months, with bootstrap 95% intervals [0.0410, 0.0416], [0.1057, 0.1067], and [0.1413, 0.1426], respectively. We also report declining risk-set support, including only 8.19% remaining at risk at 36 months. These results confirm that censoring and early repayment are quantitatively important while also supporting the narrower conclusion that outcome timing materially changes the evaluation problem. We do not claim that the mature subset represents the full origination cohort.

Changes in the revised manuscript: Expanded “Data, outcome, and information boundary” and “Chronological design and outcome maturity”; added “Right censoring and competing-risk robustness”; added the competing-risk results and cumulative-incidence table; revised the Discussion, Scope and limitations, and Conclusion to distinguish the two estimands and acknowledge selection.

Comment 2. *Model calibration and threshold selection: The selected model substantially underpredicts default risk. Recalibrate the model using an appropriate validation sample, report calibration diagnostics before and after adjustment, and reassess the threshold analysis and associated policy conclusions.*

Response: We agree. The full-training HistGB model is retained because it has the highest validation ROC-AUC among the full-training candidates (0.6966 versus 0.6932 for MLP and 0.6926 for logistic regression), without using test performance for model selection. We then fit a logistic recalibration model on the outcome-mature validation cohort only and apply the fitted transformation unchanged to the test predictions before threshold optimisation.

The revised manuscript reports mean prediction, ROC-AUC, Brier score, log loss, ten-bin ECE, calibration intercept, and calibration slope before and after recalibration. On the test cohort, ECE decreases from 0.1188 to 0.0583, Brier score from 0.1367 to 0.1232, and log loss from 0.5292 to 0.4123, while ROC-AUC remains 0.6750. Residual temporal miscalibration remains: the recalibrated mean prediction is 0.0882 compared with an observed event rate of 0.1464, with calibration intercept 0.3194 and slope 0.8722.

Threshold optimisation was then repeated using recalibrated validation probabilities. The validation-selected thresholds are 0.1730, 0.0962, and 0.0327 under 5:1, 10:1, and 20:1 FN:FP cost scenarios. On the test cohort, the corresponding review rates are 7.75%, 37.53%, and 85.06%, while FNRs are 84.35%, 41.36%, and 4.90%. The policy interpretation is therefore revised: these are analytical operating scenarios, not deployment recommendations, and any implementation would require institution-specific cost estimates, capacity constraints, consumer-outcome assessment, and continuing calibration monitoring.

Changes in the revised manuscript: Added “Validation-only probability recalibration” and “Threshold selection after recalibration”; replaced the original threshold analysis with the recalibrated analysis; added calibration and threshold tables; revised the Abstract, policy synthesis, Discussion, limitations, and Conclusion.

Comment 3. *Statistical inference: The conclusion that increased model complexity provides no meaningful performance advantage requires stronger evidence. Conduct appropriate paired statistical comparisons, report confidence intervals for performance differences, and reconsider claims of equivalence where these cannot be established.*

Response: We agree and have removed the earlier equivalence-style framing. The revised manuscript uses a paired stratified bootstrap with 2,000 replicates on the fixed 41,145-loan test cohort, resampling within adverse-outcome and non-event strata and using the same resampled observations for each model pair. At 180,000 training observations, Logistic-HistGB ROC-AUC difference is -0.0516 (95% CI [-0.0587, -0.0446]), Logistic-MLP is -0.0729 (95% CI [-0.0785, -0.0670]), and HistGB-MLP is -0.0213 (95% CI [-0.0265, -0.0159]). Under full-cohort training, the direction reverses: Logistic-HistGB is +0.0243 (95% CI [0.0184, 0.0297]), Logistic-MLP is +0.0530 (95% CI [0.0460, 0.0600]), and HistGB-MLP is +0.0287 (95% CI [0.0213, 0.0360]). The reported two-sided bootstrap p-value for each ROC-AUC contrast is 0.001.

The revised conclusion is not that the model families are equivalent. It is that statistically distinguishable differences exist, but the ordering changes materially with training-sample size, so increased representational complexity does not provide a stable out-of-time advantage across the evaluated regimes.

Changes in the revised manuscript: Added a formal paired-inference subsection and paired-bootstrap table; revised the training-size analysis, policy propositions, Abstract, Discussion, and Conclusion; removed the “no capacity premium” and equivalence-style claims.

Comment 4. *Stress-regime and subgroup findings: Several findings rely on relatively small samples and event counts. Establish defensible minimum sample requirements, report appropriate uncertainty estimates, and qualify conclusions that cannot be supported by the available evidence.*

Response: We agree. The revised analysis no longer characterises the unemployment comparison as a national macroeconomic stress regime. It is explicitly labelled a high-state-unemployment subgroup diagnostic because the variable primarily varies across states within the test period. The high-unemployment subgroup contains 641 loans and 74 adverse outcomes, compared with 40,504 loans and 5,951 adverse outcomes in the other-state group. We report bootstrap confidence intervals for subgroup ROC-AUC and Wilson 95% intervals for FNR. At the recalibrated 10:1 threshold, high-unemployment FNR is 0.5946 (95% CI [0.4808, 0.6991]) versus 0.4114 (95% CI [0.3989, 0.4239]) in the other-state group.

For grade-by-term monitoring, a cell is now reported only when both unemployment groups contain at least 10 adverse outcomes and at least 10 non-events. Only four cells satisfy this rule: C-36, C-60, D-36, and D-60 months. The high-unemployment cells contain only 11-21 adverse outcomes, so the intervals remain wide. We therefore interpret these results only as hypothesis-generating monitoring and escalation signals, not as causal evidence or definitive subgroup failure.

Changes in the revised manuscript: Reframed the unemployment analysis, added explicit minimum-support rules and uncertainty estimates, suppressed unsupported cells from inferential interpretation, and revised the subgroup table, policy synthesis, Discussion, limitations, figures, and Conclusion.

Comment 5. *Loan definitions and model specifications: Clarify the treatment of loan maturity, early repayment, renewal procedures, and default classification. Justify the selection of 180,000 training observations from the larger available sample and provide sensitivity analyses using alternative training-sample sizes.*

Response: We have clarified each of these issues. For binary classification, the adverse-outcome endpoint includes Charged Off, Default, policy-related Charged Off, Late (16-30 days), and Late (31-120 days) statuses, and is therefore broader than terminal charge-off. In the competing-risk analysis, observed charge-off is the timed adverse event and Fully Paid, including early repayment before contractual term, is treated as a competing event. No loan-renewal or refinancing process is modelled. The manuscript now also defines “review rate” explicitly as the proportion of loans whose predicted adverse-outcome risk exceeds the analytical review threshold, which is unrelated to loan renewal.

We also agree that the former 180,000-observation training design required justification. The revised analysis uses the reproducibly reconstructed full mature training cohort of 1,304,058 loans as the principal full-training condition. The 180,000 condition is retained only as a sensitivity anchor, and each model family is re-estimated on nested samples of 180,000, 360,000, 540,000, and 1,304,058 mature loans with fixed model specifications. Model ranking changes across these conditions, which is now itself a substantive governance finding.

Changes in the revised manuscript: Expanded the outcome and loan-status definitions; clarified early repayment and absence of renewal/refinancing; added a mathematical definition of review rate; documented model hyperparameters; and added the four-level training-size sensitivity analysis and table.

Comment 6. *Reproducibility: Provide the relevant analysis code and sufficient methodological documentation to permit verification of the sample construction, model estimation, calibration procedures, and reported results, in accordance with the journal's policies.*

Response: We agree and have made the analysis code publicly available. The repository reproduces sample construction, competing-risk analysis, preprocessing, model estimation, training-size sensitivity, paired inference, validation-only recalibration, threshold optimisation, and subgroup diagnostics. It also records software dependencies, random seeds, analysis parameters, and instructions for reproducing the reported tables and figures from the cited public dataset. The revised Supplementary Material additionally includes a reproducibility and policy-audit specification covering source/version information, eligibility rules, counts before and after exclusions, feature availability, preprocessing parameters, model versions, calibration sample, threshold objective, metric definitions, uncertainty procedures, and subgroup suppression rules.

Changes in the revised manuscript: Added a Code availability statement and reproducibility Supplementary Material. Public repository: https://github.com/datascintist-abusufian/Government-ai-policy-credit-decisions.

Comment 7. *Conceptual contribution: Clarify how the proposed governance framework extends existing research. Distinguish recommendations supported directly by the empirical findings from those based on regulatory guidance or conceptual reasoning.*

Response: We have sharpened the conceptual contribution and explicitly limited it. The revision does not claim that “outcome maturity” is a new statistical treatment of censoring. We now position survival and competing-risk methods as established and cite the relevant credit-scoring literature. The contribution is governance-oriented: to connect outcome-readiness evidence, calibration, threshold-conditioned decision loss, distribution shift, and sample-qualified subgroup performance to pre-specified tolerances, accountable owners, escalation actions, and reproducible decision lineage.

The manuscript now separates empirical controls from institutional design implications. Outcome-readiness diagnostics, training-size sensitivity, validation-only recalibration before probability-dependent threshold analysis, explicit threshold consequences, and minimum event support are motivated directly by the reported empirical results. Named control ownership, escalation authority, human oversight, governing-body reporting, and release or suspension decisions are identified as institutional design implications derived from the regulatory and model-risk frameworks. We state explicitly that the empirical analysis does not estimate the causal effect or legal sufficiency of any particular governance arrangement.

Changes in the revised manuscript: Revised the Introduction, “Policy interpretation,” Policy and Governance Framework, Discussion, and Conclusion; added established survival/competing-risk references; revised the governance figures and supporting text.

Comment 8. *Interpretation and conclusions: Reassess the principal claims following the additional analyses. Ensure that the abstract, discussion, and conclusions accurately reflect the strength of the empirical evidence and acknowledge any remaining limitations.*

Response: We have comprehensively revised the interpretation. The Abstract now reports the 10.3% mature-test share, training-size-dependent model ranking, paired inference, validation-only recalibration, residual temporal miscalibration, recalibrated thresholds and review burdens, and the limited support of the high-state-unemployment subgroup. The Discussion and proposition adjudication distinguish what is supported, supported

only in qualified form, and supported only as a monitoring proposition. The revised limitations explicitly acknowledge mature-cohort selection, declining time-to-event support, residual out-of-time miscalibration, small subgroup event counts, the analytical nature of the cost scenarios, unobserved rejected applicants, absence of protected-class attributes, and limited generalisability. The Conclusion no longer recommends a universal model or threshold and does not make causal or unlawful-discrimination claims.

Changes in the revised manuscript: Rewritten Abstract, policy adjudication, Discussion, Scope and limitations, and Conclusion to match the strength of the revised evidence.

Reviewer 2

Comment 1. *The study should provide further details regarding the definition of "loan maturity" and loan outcomes, including the specific accounting period used to determine whether a loan has reached maturity.*

Response: Thank you. We have now specified the accounting horizon and maturity rule in detail. Outcome maturity for the conventional classification analysis is assessed using outcome information through the April 2020 accounting cutoff. A loan is mature if it has reached a terminal status (Fully Paid, Charged Off, or Default) by that date, or if observed follow-up from origination to April 2020 is at least equal to the contractual term. Contractual terms are 36 or 60 months, so unresolved loans with shorter observable follow-up remain immature. We also distinguish the binary adverse-outcome endpoint from the competing-risk endpoint: the classification endpoint includes Charged Off, Default, policy-related Charged Off, Late (16-30 days), and Late (31-120 days), whereas the competing-risk analysis uses charge-off as the timed adverse event and full repayment as a competing event.

Changes in the revised manuscript: Expanded "Data, outcome, and information boundary," "Chronological design and outcome maturity," the sample-flow tables, and the competing-risk subsection.

Comment 2. *The calibration results indicate significant variation in renewal rates. Consequently, the authors should discuss this variation, supported by additional analysis ideally including a technical description of the renewal process and the specific renewal methods employed.*

Response: We appreciate this comment and believe the reference to "renewal rates" concerns the manuscript's review rates. The study does not model loan renewal, refinancing, or extension, and no renewal process or renewal method is estimated. To remove the ambiguity, the revised manuscript now defines review rate mathematically as R(t) = N^{-1} sum I(p_i >= t), namely the proportion of loans whose calibrated predicted adverse-outcome risk exceeds the operating threshold and would be flagged for analytical review.

We also repeated the threshold analysis after validation-only recalibration. The large variation in review rate is explained by the cost-sensitive operating rule: 5:1, 10:1, and 20:1 FN:FP scenarios select thresholds of 0.1730, 0.0962, and 0.0327, yielding test review rates of 7.75%, 37.53%, and 85.06%, respectively. Thus, the variation is a consequence of the assumed decision costs and selected threshold, not a loan-renewal mechanism. We have stated this explicitly and have avoided using "renewal" for this quantity.

Changes in the revised manuscript: Added the technical definition of review rate and explicit statement that it is unrelated to loan renewal/refinancing/extension; revised the recalibrated threshold analysis and its interpretation.

Reviewer 3

Novelty and Conceptual Contribution

Comment. *The governance issue is real, but "outcome maturity" recasts the concepts of right-censoring and competing risks in the survival analysis credit scoring literature without proper citation of them. The audit significantly advances the model-risk agenda in that it connects calibration, thresholding, and slicing to regulatory mechanisms, although each analytical component has been thoroughly developed separately. The results have relevance for the model risk and regulatory communities, but only if the estimands issues are solved. The governance equation is presented as A_t = g(O_t, C_t, D_t, S_t, W_t), but several of the*

components and the breach term B_t are ill-defined.

Response: We agree with this conceptual clarification. The revised manuscript now explicitly states that outcome maturity is not proposed as a new survival-analysis concept or statistical solution to censoring. We added established credit-risk survival and competing-risk references, including Stepanova and Thomas (2002), Bellotti and Crook (2009), Dirick et al. (2017), and Li et al. (2023). The mature-cohort classification estimand is now separated from a complementary competing-risk estimand that retains unresolved loans as right-censored observations and treats full repayment as a competing event.

We have also narrowed the novelty claim. The contribution is the governance integration of established diagnostics into an evidence-to-control process in which outcome readiness, calibration error, threshold-conditioned decision loss, distribution-shift evidence, and worst supported subgroup/slice loss are tied to institution-approved tolerances and owned actions. The revised framework defines O_t as outcome readiness; C_t as calibration error; D_t as threshold-conditioned decision loss; S_t as a pre-specified distribution-shift statistic; and W_t as the worst supported subgroup or slice-level decision loss. Larger C_t, D_t, S_t, and W_t indicate greater risk, while greater O_t indicates greater readiness. The breach term is now explicitly defined as $B_t = I[O_t < O^* \text{ or } C_t > C^* \text{ or } D_t > D^* \text{ or } S_t > S^* \text{ or } W_t > W^*]$. The starred values are institution-approved tolerances set ex ante, not universal regulatory thresholds, and g is a documented institutional mapping from evidence and breach severity to actions such as continue, recalibrate, human review, restrict automation, or suspend. Neither g nor the tolerances are estimated by our retrospective analysis.

Changes in the revised manuscript: Revised the Introduction, survival/competing-risk literature discussion, Policy and Governance Framework, governance equation and breach definition, figures, Discussion, and Conclusion.

Methodological Rigor and Robustness

Comment. *The reviewer raises concerns that the 10.3% outcome-mature test subset may be unrepresentative; that HistGB was selected by validation ROC-AUC despite different test rankings and poor calibration; that the earlier "no capacity premium" claim was based on small seed dispersion rather than paired testing; that the 180,000-loan training sample was not justified; and that replication was weak because code was not publicly available.*

Response: We have addressed each component. First, we now describe the mature-cohort result as a conditional estimand and add a competing-risk analysis that retains unresolved observations as right-censored rather than treating the mature subset as representative of all originations. Second, model selection is explicitly based only on the pre-specified validation criterion: under full training HistGB has validation ROC-AUC 0.6966, compared with 0.6932 for MLP and 0.6926 for logistic regression. Test results are reported for evaluation and are not used retrospectively for model selection. Because the selected HistGB probabilities are miscalibrated, we fit validation-only logistic recalibration and repeat all probability-dependent threshold analyses after recalibration.

Third, the earlier seed-dispersion argument and "no capacity premium" language have been replaced by paired stratified bootstrap inference on the common test cohort, with 95% confidence intervals and p-values. Fourth, the former 180,000-observation design is no longer treated as a preferred training sample. We re-estimate all model families using nested mature training samples of 180,000, 360,000, 540,000, and the full 1,304,058-loan cohort while holding model specifications fixed. The resulting ranking reversal is reported directly. Finally, the analysis code and reproducibility instructions are now public.

Changes in the revised manuscript: Revised "Models, selection, and uncertainty," added training-size and paired-inference subsections/tables, added validation-only recalibration, and added the public Code availability statement and Supplementary reproducibility specification.

Major Concern 1

Comment. *The outcome maturity condition will definitely induce an unknown form of bias since the loans that have been considered here belong to early terminating loans due to median test group maturity period*

of about four months compared to 36 to 60 month periods. Thus, the 14.64% default rate will definitely be improperly bounded. Use survival/competing risk analysis.

Response: We agree with the core concern and no longer interpret the 14.64% mature-cohort rate as an unbiased estimate for all test-period originations. The revision explicitly states that maturity restriction may preferentially retain loans that resolve earlier and changes the estimand. We therefore added the requested competing-risk analysis over the full time-to-event cohort. Unresolved loans remain right-censored, full repayment is treated as a competing event, and charge-off cumulative incidence is reported at supported horizons. The analysis includes 2,700,690 loans with usable timing, of which 1,760,207 are right-censored, 765,339 fully paid, and 175,144 charged off. We also report risk-set support and bootstrap intervals and avoid extrapolating beyond the observed support. The classification and competing-risk analyses are explicitly presented as related but non-interchangeable estimands.

Changes in the revised manuscript: Added the competing-risk Methods and Results sections and table; revised the maturity interpretation, Discussion, limitations, and Conclusion.

Major Concern 2

Comment. *The thresholds and cost-sensitive conclusions were drawn using an uncalibrated model selected on validation ROC-AUC. Recalibrate the model prior to the choice of thresholds and redo all subsequent analysis.*

Response: We have implemented this request. Model selection remains validation-only, after which a logistic recalibration model is fitted exclusively on the mature validation cohort and applied unchanged to test predictions. We report pre- and post-recalibration diagnostics. On test, ECE improves from 0.1188 to 0.0583, Brier score from 0.1367 to 0.1232, and log loss from 0.5292 to 0.4123, while ROC-AUC remains 0.6750. Residual underprediction remains and is explicitly acknowledged.

All threshold optimisation and subsequent subgroup decision diagnostics were then recomputed on the recalibrated probability scale. The revised thresholds are 0.1730, 0.0962, and 0.0327 for the 5:1, 10:1, and 20:1 cost scenarios, with test review rates of 7.75%, 37.53%, and 85.06% and FNRs of 84.35%, 41.36%, and 4.90%, respectively. These scenarios are now interpreted as analytical policy illustrations rather than deployment recommendations.

Changes in the revised manuscript: Added recalibration Methods and Results, replaced the threshold table and downstream threshold analysis, and revised all associated policy claims.

Major Concern 3

Comment. *The "no capacity premium" claim relies on a small AUC spread with no paired significance test. Report paired bootstrap/DeLong tests for model comparisons.*

Response: We agree. We replaced the earlier spread-based argument with paired stratified bootstrap comparisons using 2,000 replicates on the common test cohort. Confidence intervals and two-sided bootstrap p-values are now reported for each ROC-AUC contrast. The direction of the statistically distinguishable differences reverses between the 180,000-observation and full-training conditions. We therefore removed the "no capacity premium" and equivalence-style claims. The revised claim is that model ordering is training-size dependent and that increased representational complexity does not provide a stable out-of-time advantage across the evaluated training regimes.

Changes in the revised manuscript: Added the paired-bootstrap Methods and Results subsection and table; revised the Abstract, proposition adjudication, Discussion, and Conclusion.

Major Concern 4

Comment. *The complete miss detection and stress regime claims are over-promoted beyond their statistical merit (n=7 and n=641 respectively), and the normative controls of the governance framework are not*

distinguished empirically from verified claims. Relabel these as research hypothesis generation alarms, and set out minimum event frequency requirements.

Response: We agree. Sparse cell claims have been removed from operative inference. The unemployment analysis is now described as a cross-sectional high-state-unemployment subgroup diagnostic, not a national stress regime. A grade-by-term cell is reported only when both unemployment groups contain at least 10 adverse outcomes and at least 10 non-events. Only four cells satisfy this rule, and their high-unemployment event counts remain 11-21, so we explicitly characterise the evidence as hypothesis-generating monitoring and escalation signals rather than causal or definitive subgroup effects. Unsupported cells are suppressed from directional interpretation.

We also now distinguish empirically motivated controls from normative institutional design. The empirical analysis supports controls relating to outcome readiness, training-size sensitivity, recalibration, threshold consequences, and minimum support. Recommendations about named ownership, escalation authority, human oversight, governing-body reporting, and release/suspension decisions are identified as regulatory and conceptual design implications rather than empirically estimated effects.

Changes in the revised manuscript: Revised subgroup Methods and Results, introduced explicit support rules and uncertainty, revised Policy interpretation, Discussion, Scope and limitations, governance figures, and Conclusion.

Tables, Appendix, and Figures

Comment. *We noticed that in-text citation for table 13 is missing in the main manuscript. Please make the required changes to proceed. Please maintain uniformity while labelling and citing Tables, Appendix and Figure files. Please ensure that tables, appendix and figures are correctly cited in the manuscript.*

Response: Thank you. We audited the revised manuscript for cross-references and standardised table, figure, and supplementary-material labels. The table structure changed substantially during revision, so the former numbering is not preserved one-to-one. Every table retained in the revised manuscript is now introduced or cited in the text, including the sample-flow, maturity, training-size, paired-bootstrap, calibration, threshold, supported-slice, competing-risk, policy-synthesis, provenance, governance-control, and audit-package tables. The model-governance architecture, evidence-to-control framework, and institutional-control architecture are also cited in the main text. The Supplementary reproducibility section and its audit-package table are explicitly referenced from the Code availability statement. We have also revised the figure files and captions to use terminology consistent with the revised analysis.

Changes in the revised manuscript: Completed a manuscript-wide cross-reference audit and standardised table, figure, and Supplementary Material references and labels.

Closing statement

We thank the editor and reviewers again for the detailed comments. The revision has materially changed both the empirical analysis and the scope of the claims. The manuscript now treats outcome maturity as a governance-readiness condition rather than a substitute for survival analysis; separates mature-cohort classification from competing-risk estimation; performs training-size sensitivity and paired inference; recalibrates before threshold selection; imposes sample-qualified subgroup reporting; distinguishes analytical review rates from loan renewal; provides public code and reproducibility documentation; and limits the governance conclusions to what the empirical evidence and cited regulatory frameworks can support. We hope these revisions satisfactorily address the methodological and conceptual concerns raised.

Public analysis repository: https://github.com/datascintist-abusufian/Government-ai-policy-credit-decisions